

Jimmy Choo, CFA, works for Joel Constable, a high net worth investor who invests a large proportion of his portfolio speculatively in order to chase high returns. Choo is usually tasked with carrying out initial analysis on investment opportunities, which Constable has identified and may proceed further with.

Most of the opportunities identified by Constable are investments in private companies with little history of profits or dividends. Some, however, are more stable growth companies, which Constable still believes to be undervalued.

One opportunity recently identified by Constable is a private company, Bakan Fashion. Bakan is a fashion retailer, which started up recently with an online presence and a retail unit in a high-end location on a city high street. The company began trading four years ago and has published the following financial information for the benefit of its investors:

Bakan Fashion

Income Statement (\$m) (extracts)				
	2x07	2x08	2x09	2x10
Revenue	0.3	0.8	1.9	1.9
Net income	(0.9)	(0.9)	(0.3)	0.1
Cash flow statement (\$m) (extracts)				
	2x07	2x08	2x09	2x10
CFO	(0.6)	(0.7)	(0.3)	0.2
CFI	(3.6)	(3.2)	(1.0)	(0.4)
CFF	5.6	1.0	0.4	0.4
Balance sheet (\$m) (extracts)				
	2x07	2x08	2x09	2x10
Net assets	3.2	3.0	2.6	2.9

Choo has also made some additional notes regarding the 2x10 net income figure after meeting briefly with the finance director of Bakan.

Notes Re: 2x10 Net Income

- CEO compensation package is \$650,000 per annum
- A leading fashion supplier provided the retail outlet at a rate of \$120,000 per annum for the first four years
- A friend of the CEO helped set up the online operations for a discounted cost of \$20,000
- The CEO agreed to the compensation package in a bid to help the company breakeven as quickly as possible. Choo estimates that a market based compensation figure would be \$850,000
- The market lease rate for the retail unit is \$170,000
- The set up cost for the online operation should have been \$50,000

In addition to the financial information for Bakan, Choo has also identified the following specific factors, which he always takes into account in any analysis of private companies for Constable:

Private company equity typically has fewer potential owners and is less liquid
Factor 1: than publicly traded shares; hence a discount should be applied in any valuation of Bakan.

One key objective of the board of Bakan may be to limit exposure to taxes, as
Factor 2: 75% of the board members also own substantial equity in the company.

Due to the significant ownership of equity by the board of Bakan, management
Factor 3: may be tempted to take a shorter-term view than that typically taken by public firms.

Constable always prefers to see a basic analysis of price multiples for companies before any valuation.

Choo produces this where possible, but usually follows it with a valuation of the private company-using price multiples based on recent sales of comparable assets. He calls this the "Choo Comparison Method (CCM)."

The CCM valuation is then adjusted by Constable and Choo to take account of Constable's individual financing costs and any perceived synergies with his existing assets.

Choo refers to this final valuation after adjustments in his reports as the "Intrinsic Value." Constable has recently taken him to task on this and suggested that as the valuation focuses on the value to him as a specific investor it should be called the "Investment Value."

Question #1 - 6 of 48

Question ID: 1586227

Assuming Choo uses price multiples based on the four years of financial information from Bakan, ignoring his additional notes regarding 2x10, which of the following statements is least accurate?

- A) Price to Earnings ratios would be meaningless.
 - B) Price to Book ratios would be the most useful as they would take account of the human capital in a start-up business such as Bakan.
 - C) Price to Cash Flow ratios are theoretically better if calculated using free cash flow to equity rather than CFO.
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Question #2 - 6 of 48

Question ID: 1473351

When valuing private companies, how would the factors described by Choo usually be classified?

- A) Factor 1 Stock-specific, Factor 2 Stock-specific.
 - B) Factor 1 Stock-specific, Factor 2 Company-specific.
 - C) Factor 1 Company-specific, Factor 2 Company-specific.
-

Question #3 - 6 of 48

Question ID: 1473352

Is Choo correct in his assessment of Factor 3?

- A) Yes.
 - B) No, it is rare for private companies to allow key management to own equity.
 - C) No, board equity ownership usually results in a longer-term view than public companies.
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Question #4 - 6 of 48

Question ID: 1586228

The "Choo Comparison Method" of valuing a private company is consistent with which major approach?

- A)** Income Approach.
 - B)** Multiple Approach.
 - C)** Market Approach.
-

Question #5 - 6 of 48

Question ID: 1586229

Is Constable right in his suggested renaming of the final valuation?

- A)** Yes.
 - B)** No, a valuation based on analysis of company fundamentals should be referred to as an Intrinsic Value.
 - C)** No, it should be renamed Market Value as it uses primarily market information.
-

Question #6 - 6 of 48

Question ID: 1586230

Using Choo's additional notes on Bakan's 2x10 Net Income, which of the following is the most accurate estimate of normalized earnings for 2x10?

- A)** Loss \$150,000.
 - B)** Profit \$250,000.
 - C)** Loss \$180,000.
-

Question #7 of 48

Question ID: 1473310

An analyst is examining three companies. Given the information below, which of them is *most likely* to be a private firm?

Firm	Number of Years in Operation	Market Capitalization	Required Return for Common Stock
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A	12 years	\$1,324.8 million	14.8%
B	4 years	\$1,313.9 million	18.3%
C	19 years	\$2,231.0 million	16.4%

A) Firm A.

B) Firm C.

C) Firm B.

Question #8 of 48

Question ID: 1586192

Given the following figures, calculate the FCFF. Assume the earnings and expenses are normalized and that capital expenditures will cover depreciation plus 3 percent of the firm's incremental revenues.

<i>Current Revenues</i>	\$30,000,000
<i>Revenue growth</i>	6%
<i>Gross profit margin</i>	20%
<i>Depreciation expense as a percent of sales</i>	1%
<i>Working capital as a percent of sales</i>	15%
<i>SG&A expenses</i>	\$3,800,000
<i>Tax rate</i>	30%

A) \$927,400.

B) \$1,245,400.

C) \$1,785,400.

Question #9 of 48

Question ID: 1586199

Using the following information, calculate the required return on equity using the expanded CAPM.

<i>Income return on bonds</i>	6.0%
<i>Capital return on bonds</i>	2.0%
<i>Long-term Treasury yield</i>	3.5%
<i>Beta</i>	1.4
<i>Equity risk premium</i>	6.0%
<i>Small stock premium</i>	4.0%
<i>Company-specific risk premium</i>	3.0%
<i>Industry risk-premium</i>	2.0%
<i>Pretax cost of debt</i>	11.0%
<i>Optimal Debt/Total Cap</i>	16%
<i>Current Debt/Total</i>	7%
<i>Debt/Total Cap for public firms in industry</i>	33%
<i>Tax Rate</i>	30%

- A) 11.9%.
- B) 18.9%.
- C) 15.9%.

Using the following figures, calculate the value of the firm using the excess earnings method (EEM).

<i>Working capital</i>	\$600,000
<i>Fixed assets</i>	\$2,300,000
<i>Normalized earnings</i>	\$340,000
<i>Required return for working capital</i>	5%
<i>Required return for fixed assets</i>	13%
<i>Growth rate of residual income</i>	4%
<i>Discount rate for intangible assets</i>	18%

- A) \$2,981,714.
 - B) \$3,073,199.
 - C) \$3,027,111.
-

Question #11 of 48

Question ID: 1473312

Which of the following is *least likely* an example of a compliance-related valuation for a private company?

- A) Financial reporting.
 - B) Bankruptcy proceeding.
 - C) Tax purposes.
-

Question #12 of 48

Question ID: 1586211

An analyst is valuing a small private firm that is still developing and has yet to generate any earnings. Which of the following *best* describes the approach that should be used?

- A) An asset-based approach would be used.
- B) A market approach based on public comparables would be utilized.

- C)** Nonoperating assets are not crucial to the firm and should be excluded in any valuation.
-

Question #13 of 48

Question ID: 1586193

Which of the following *best* describes how debt is incorporated into the estimation of the discount rate for private company valuations, relative to that for public firms? In general, the cost of debt:

- A)** and debt capacity is the same for both private and public firms.
 - B)** is higher for private firms and debt capacity is lower for private firms.
 - C)** is higher for private firms and debt capacity is the same for both private and public firms.
-

Question #14 of 48

Question ID: 1586197

Which of the following *best* describes the build-up method used for the estimation of the discount rate in private company valuations?

- A)** It is useful when there are no comparable public firms.
 - B)** An industry risk premium is not included because it is captured in the equity risk premium.
 - C)** Because it is not used in the calculation, beta is assumed to be zero.
-

Question #15 of 48

Question ID: 1473316

A private business is being valued for the purpose of determining the appropriate level of performance-based managerial compensation. This private company valuation would be *best* described as a:

- A)** Litigation-related valuation
- B)** Transaction-related valuation
- C)** Compliance-related valuation

Question #16 of 48

Question ID: 1586213

Which of the following is *most* accurate regarding the asset-based approach? Of the three valuation methods for private firms, it usually:

- A)** is the most appropriate for going concerns.
 - B)** results in the lowest valuation.
 - C)** is not difficult to apply.
-

Question #17 of 48

Question ID: 1586215

The capitalized cash flow method (CCM) used in private firm valuation is *most* appropriate when:

- A)** earnings are growing quickly in an initial period.
 - B)** stable growth is expected.
 - C)** there are many intangible assets to value.
-

Question #18 of 48

Question ID: 1586209

Which of the following approaches to private company valuation uses discounted cash flow analysis?

- A)** The market approach.
 - B)** The income approach.
 - C)** The asset-based approach.
-

Question #19 of 48

Question ID: 1586214

When would the asset-based approach result in a higher valuation than its going concern value, in the case of private company valuation?

- A) If the firm has minimal profits and poor prospects.
 - B) When valuing pharmaceutical firms.
 - C) When valuing biotech firms.
-

Question #20 of 48

Question ID: 1586194

Which of the following *best* describes the use of size premiums when estimating the discount rate for private company valuations?

- A) The treatment is similar to that for public firms.
 - B) A size premium is subtracted when calculating the discount rate.
 - C) When using data from comparable public firms, a distress premium may be inadvertently added in.
-

Question #21 of 48

Question ID: 1586203

Assume a minority shareholder holds 10% of a private firm's equity, with the CEO holding the other 90%. Using normalized earnings, the value of the firm's equity is estimated at \$20 million. The CEO refuses to sell the firm and the minority shareholder cannot sell their interest easily. A discount for lack of marketability (DLOM) of 15% will be applied. A discount for lack of control (DLOC) will also be estimated. Using reported earnings instead of normalized earnings provides an estimated firm equity value of \$19 million. Which of the following is *closest* to the value of the minority shareholder's equity interest?

- A) \$1,700,000.
 - B) \$1,900,000.
 - C) \$1,615,000.
-

Question #22 of 48

Question ID: 1586190

Which of the following *best* describes the use of FCFF and FCFE when used in private firm valuation?

- A) FCFE is usually favored if the firm is going to change its capital structure because the equityholders are usually the investors requesting the valuation.
 - B) FCFE is usually favored if the firm is going to change its capital structure because the cost of equity is less sensitive to leverage changes than the WACC.
 - C) FCFF is usually favored if the firm is going to change its capital structure because the WACC is less sensitive to leverage changes than the cost of equity.
-

Paul Smith is an analyst performing valuations for Lumber Limited. Smith has been given a project to value Timber Industries, a firm that Lumber Limited is considering acquiring. Smith is aware that a number of characteristics distinguish private and public companies, and that these characteristics must be considered during his process of valuing Timber Industries. A number of issues complicate Smith's valuation: Timber Industries pays its CEO well below a market-based compensation figure, leases a warehouse at an above-market rate, and owns a vacant office building that is not needed for core operations. Smith is also aware that discounts and premiums based on control and marketability must be considered in his valuation of Timber Industries.

Question #23 - 26 of 48

Question ID: 1586219

Compared to a public company, it is *most likely* that as a private company Timber Industries will have greater:

- A) focus on the short-term.
 - B) quality and depth of management.
 - C) concerns related to taxes.
-

Question #24 - 26 of 48

Question ID: 1586220

One valuation method that Smith is considering for Timber Industries involves using a growing perpetuity formula to estimate the value of intangible assets, and then adding this value to the values of working capital and fixed assets. This method is *most accurately* described as the:

- A) free cash flow method.
- B) excess earnings method.

C) capitalized cash flow method.

Question #25 - 26 of 48

Question ID: 1586221

The asset-based approach to private company valuation that Smith is considering for Timber Industries is *most likely* to be appropriate in the case of a:

- A) finance firm such as a bank.
 - B) mature company with many intangible assets.
 - C) firm with strong profits and growth potential.
-

Question #26 - 26 of 48

Question ID: 1586222

Which of the following statements related to discounts and premiums to benchmark for Smith's private company valuation of Timber Industries is *most accurate*?

- A discount for lack of marketability should be applied when the comparables are
- A) based on public shares, and the interest in the target company is a minority interest in a private firm.
- A discount for lack of control should be applied when the comparable company
- B) values are for public shares, and the target company valuation is for a controlling interest.
- A control premium should be added when the comparable values are for the sale of
- C) an entire company, and the valuation is being done for a minority interest in the target company.
-

Question #27 of 48

Question ID: 1586191

Given the following figures, calculate the normalized EBITDA for a financial and strategic buyer.

<i>Reported EBITDA</i>	\$4,500,000
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<i>Current Executive Compensation</i>	\$700,000
<i>Market-Based Executive Compensation</i>	\$620,000
<i>Current SG&A expenses</i>	\$6,300,000
<i>SG&A expenses after synergistic savings</i>	\$5,600,000
<i>Current Lease Rate</i>	\$300,000
<i>Market-Based Lease Rate</i>	\$390,000

The normalized EBITDA for each type of buyer is:

Financial Buyer Strategic Buyer

- A)** \$4,490,000 \$5,190,000
- B)** \$4,190,000 \$4,890,000
- C)** \$4,670,000 \$5,370,000
-

Using the following information, calculate the WACC using the build-up method, assuming the firm is being acquired.

<i>Income return on bonds</i>	6.0%
<i>Capital return on bonds</i>	2.0%
<i>Long-term Treasury yield</i>	3.5%
<i>Beta</i>	1.4
<i>Equity risk premium</i>	6.0%
<i>Small stock premium</i>	4.0%
<i>Company-specific risk premium</i>	3.0%
<i>Industry risk-premium</i>	2.0%
<i>Pretax cost of debt</i>	11.0%
<i>Optimal Debt/Total Cap</i>	20%
<i>Current Debt/Total</i>	7%
<i>Debt/Total Cap for public firms in industry</i>	33%
<i>Tax Rate</i>	30%

- A) 18.5%.
- B) 16.3%.
- C) 17.7%.

Question #29 of 48

Question ID: 1586212

The asset-based approach values a firm based on:

- A) investment values.
 - B) fair values.
 - C) book values.
-

Question #30 of 48

Question ID: 1586196

A private pharmaceutical firm is under consideration for acquisition where the financial buyer will pay with equity. Part of the payment to the sellers is based on FDA approval of the firm's drug. If the analyst uses a market approach and comparable data from public firms, which of the following would *most likely* result in a price-multiple that is too high? The comparable data is:

- A)** from transactions where the buyer used cash.
 - B)** for strategic buyers.
 - C)** for transactions where the consideration was non-contingent.
-

Question #31 of 48

Question ID: 1586225

An analyst is valuing a private firm on the behalf of a strategic buyer and deflates the average public company multiple by 15% to account for the higher risk of the private firm. Given the following figures, calculate the value of firm equity using the guideline public company method (GPCM).

Market value of debt	\$4,100,000
Normalized EBITDA	\$42,800,000
Average EV/EBITDA multiple	8.5
Control premium from past transaction	25%

The enterprise value of the firm is *closest* to:

- A)** \$386,484,000.
 - B)** \$304,060,000.
 - C)** \$382,384,000.
-

Question #32 of 48

Question ID: 1586201

An analyst calculates a control premium of 15% and discount for lack of marketability (DLOM) of 20%. Which of the following is *closest* to the total discount for valuing minority equity interests in the private firm?

- A) 35.7%.
 - B) 30.4%.
 - C) 35.0%.
-

Question #33 of 48

Question ID: 1473314

Which of the following is *least likely* an example of a litigation-related valuation for a private company?

- A) Lost profits claims.
 - B) Bankruptcy proceeding.
 - C) Divorce settlements.
-

Question #34 of 48

Question ID: 1586198

Which of the following statements related to the models used to estimate the required rate of return to private company equity is *most accurate*:

- A) The CAPM model uses betas estimated from firm returns of other private firms.
 - B) The expanded CAPM model adds premiums for size and firm-specific risk.
 - C) The build-up method begins with betas for comparable public firms and adds risk premiums.
-

Question #35 of 48

Question ID: 1586195

Which of the following *best* describes projection risk in the estimation of the discount rate for private company valuations?

- A) Projection risk results in higher discount rates.

B) Management will always be overly optimistic to increase the acquisition price.

C) If the availability of information from private firms is poor, the uncertainty of projected cash flows may increase.

Question #36 of 48

Question ID: 1473309

Which of the following statements *most* accurately describes the difference between private and public firm managers?

Although managers in a public firm are often paid with incentive compensation,
A) public managers may take a shorter term view than private managers because shareholders often focus on the short-term.

Because managers in a public firm are often paid with incentive compensation,
B) public managers may take a longer term view than private managers.

Because managers in a private firm are concerned with having the firm go public,
C) private managers may take a shorter term view than public managers.

Question #37 of 48

Question ID: 1586208

An analyst is valuing a firm's equity using the price-to-book-value ratio of similar firms.

Which of the following is the *most likely* valuation approach the analyst will use?

A) The income approach.

B) The market approach.

C) The asset-based approach.

Stan Bowles works for Marsh Inc. and has been tasked with the valuation of Park Limited, a small private footwear producer. Bowles prepares a valuation report on Park Limited and his report contains the following:

Company-specific characteristics such as the quality and depth of

Comment 1: management, tax considerations, and shareholders agreements that restrict liquidity mark the main differences between a private and public company.

The value of a private company depends on the investor's expectations and
Comment 2: investment requirements and could differ from one buyer to the next due to different perception of risk and future potential.

To obtain an appropriate discount rate for Park, we have assumed the following:

- Comment 3:
- Estimated beta of Park: 0.75
 - Company specific risk premium: 1.2%
 - Small stock premium: 2.8%
 - Risk free rate: 3%
 - Equity risk premium: 4.5%

Comment 4: If we are valuing Park for non-controlling equity interest, a discount for lack of control might be required.

Question #38 - 41 of 48

Question ID: 1473357

Which of the following mentioned in Comment 1 is *least* likely to be a company specific characteristic of a private company?

- A)** Tax consideration.
 - B)** Quality and depth of management.
 - C)** Shareholders agreements that restrict liquidity.
-

Question #39 - 41 of 48

Question ID: 1586205

Comment 2 is describing:

- A)** fair value.
 - B)** investment value.
 - C)** fair market value.
-

Question #40 - 41 of 48

Question ID: 1586206

Calculate Park's discount rate using the build-up method and the information in Comment 3.

- A) 9.375%.
 - B) 10.375%.
 - C) 11.5%.
-

Question #41 - 41 of 48

Question ID: 1586207

Using Comment 4, which of the following method is *least* likely to require a discount for lack of control?

- A) Guideline Transaction Method.
 - B) Capitalized Cash Flow Method.
 - C) Guideline Public Company Method.
-

Question #42 of 48

Question ID: 1586202

Which of the following *best* describes the estimation of discounts for lack of marketability (DLOM) in private company valuations? The primary advantage of using put prices to estimate the DLOM over the other two methods is:

- A) the volatility of the firm can be incorporated into the analysis.
 - B) the Black-Scholes model has been shown to be valid for private firms.
 - C) exchange traded put prices are readily available.
-

Question #43 of 48

Question ID: 1586224

Which of the following statements related to the market approaches to private company valuation is *most accurate*?

- A) The guideline public company method (GPCM) is based on price multiples from comparable traded firms.

- B)** The prior transaction method (PTM) is based on price multiples from the sale of whole public and private companies.
- C)** The guideline transactions method (GTM) is based on historical stock sales of the actual subject company.
-

Question #44 of 48

Question ID: 1473313

Which of the following is *least likely* an example of a transaction-related valuation for a private company?

- A)** Bankruptcy proceeding.
- B)** Performance-based managerial compensation.
- C)** Financial reporting.
-

Question #45 of 48

Question ID: 1473311

An analyst is examining the stock of three companies. Given the information below, which of them is *most likely* to be the stock of a private firm?

<i>Firm</i>	<i>Restrictions on Sale of Stock?</i>	<i>DLOM</i>	<i>Stock Ownership of 5 Largest Owners</i>
A	Yes	0%	28%
B	No	5%	35%
C	Yes	15%	64%

- A)** Firm A.
- B)** Firm C.
- C)** Firm B.
-

Question #46 of 48

Question ID: 1586223

An analyst values a private firm by using price multiples from the sale of whole companies, with adjustments for risk differences. Which of the following best describes the valuation method that the analyst is using?

- A)** The guideline transactions method.
 - B)** The guideline public company method.
 - C)** The prior transaction method.
-

Question #47 of 48

Question ID: 1586216

Using the following figures, calculate the value of the equity using the capitalized cash flow method (CCM), assuming the firm will be acquired.

<i>Normalized FCFE in current year</i>	\$3,000,000
<i>Reported FCFE in current year</i>	\$2,400,000
<i>Growth rate of FCFE</i>	7.0%
<i>Equity discount rate</i>	16.0%
<i>WACC</i>	13.0%
<i>Risk-free rate</i>	3.5%
<i>Cost of debt</i>	10.5%
<i>Market value of debt</i>	\$3,000,000

The value of the equity is:

- A)** \$35,666,667.
 - B)** \$32,666,667.
 - C)** \$28,533,333.
-

Question #48 of 48

Question ID: 1586210

An analyst values a private company using a price multiple based on recent sales of comparable assets. This approach to private company valuation is *best* described as the:

- A)** income approach
- B)** market approach
- C)** asset-based approach